3 A stretched string is attached to two fixed points and a vibration generator as shown in Figure 3.1.

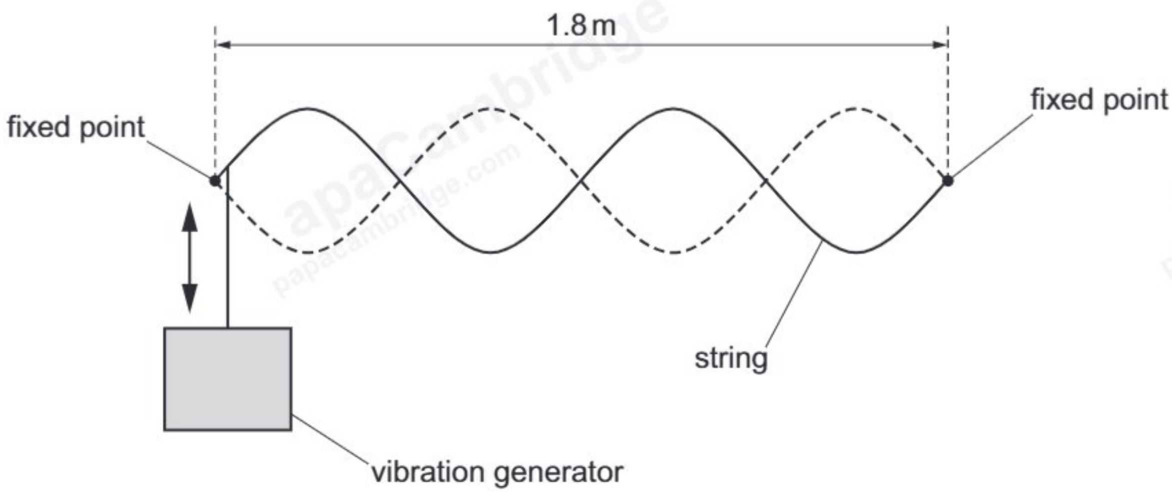

Figure 3.1

The vibration generator oscillates vertically and a stationary wave is formed on the string.

(a) Explain how the stationary wave is formed on the string.

....

....

....

....

.... [2]

(b) On Figure 3.1, draw **one** cross (×) at the position of a node. [1]

(c) The distance between the fixed points is 1.8 m. The vibration generator oscillates at a frequency of 26 Hz.

Calculate the speed of progressive waves on the string.

speed = ... $m\,s^{-1}$ [3]

(d) The frequency of the vibration generator is now halved and a new stationary wave is formed on the string. The speed of progressive waves on the string is unchanged.

On Figure 3.2, draw the string at an instant when it has maximum displacement.

Figure 3.2

[2]

[Total: 8]